

Solution Set: Module 2 Quiz

1. In the context of a simple linear regression model, what does the trade-off between bias and variance imply?

Answer: B) Minimizing bias often increases variance

Explanation: There is a fundamental trade-off between bias and variance in modeling. Minimizing bias (getting closer to the true function) often increases variance (sensitivity to changes in the dataset), especially in flexible models.

2. How does increasing model flexibility affect interpretability?

Answer: B) It decreases interpretability

Explanation: As models become more flexible (e.g., higher-order polynomials or more complex models), they tend to overfit the data, which makes them harder to interpret because their decisions are driven by more complex rules.

3. What is the primary objective of constraining or shrinking estimating coefficients in regression models?

Answer: C) To decrease variance while accepting a slight increase in bias

Explanation: Constraining or shrinking coefficients (e.g., through Ridge or Lasso regression) reduces model complexity, which decreases variance. This is done at the cost of slightly increasing bias, improving model generalization.

4. Which regression technique introduces a penalty term to keep coefficient estimates small?

Answer: B) Ridge Regression

Explanation: Ridge Regression introduces an L_2 -norm penalty, shrinking the magnitude of the regression coefficients to control model complexity and reduce overfitting.

5. What is the primary purpose of Lasso Regression?

Answer: D) To perform variable selection

Explanation: Lasso Regression uses an L_1 -norm penalty that tends to drive some coefficients to exactly zero, effectively selecting a subset of predictors, which is useful for variable selection.

6. How does the computational complexity of Ridge Regression compare to ordinary linear regression?

Answer: A) Ridge Regression is always more computationally complex

Explanation: Ridge Regression involves solving an augmented least-squares problem that includes the penalty term λ , which increases the computational complexity compared to ordinary linear regression.

7. In scenarios with a large number of features, which optimization technique is commonly used in Ridge Regression?

Answer: A) Gradient descent

Explanation: For large datasets, gradient descent is often used in Ridge Regression to efficiently find the optimal coefficients, especially when the number of features is large.

8. What is the geometric interpretation of Ridge Regression when the number of features is 2?

Answer: D) Ridge Regression has a regularization term that shrinks the coefficient estimates towards zero

Explanation: Ridge Regression geometrically shrinks the coefficient estimates by imposing an L_2 -norm constraint, pulling the coefficients closer to zero, which helps prevent overfitting.

9. Which regression technique is known for its ability to handle multicollinearity by encouraging sparse coefficient estimates?

Answer: C) Lasso Regression

Explanation: Lasso Regression addresses multicollinearity by shrinking some coefficients to exactly zero, encouraging sparsity, which helps when features are correlated (multicollinearity).

10. When selecting variables in a regression model, which technique is preferred for creating sparsity?

Answer: B) Lasso Regression

Explanation: Lasso Regression is preferred for variable selection because its L_1 -norm penalty forces some coefficients to be exactly zero, creating a sparse solution.

11. Compute the solution of the Ridge Regression model

Given:

$$X = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}, \quad y = \begin{pmatrix} 36 \\ 62 \end{pmatrix}, \quad \lambda = 3$$

We want to solve the Ridge Regression problem:

$$\min_{\beta} (\|y - X\beta\|_2^2 + \lambda\|\beta\|_2^2)$$

The solution of Ridge Regression is derived as follows:

We want to minimize the following expression:

$$L(\beta) = (y - X\beta)^T(y - X\beta) + \lambda\beta^T\beta$$

The gradient of $L(\beta)$ with respect to β is:

$$\nabla_{\beta} L(\beta) = -2X^T y + 2X^T X \beta + 2\lambda \beta$$

Set the gradient equal to zero to find the optimal β :

$$X^T X \beta + \lambda \beta = X^T y$$

Factor out β :

$$(X^T X + \lambda I) \beta = X^T y$$

Finally, solving for β gives the Ridge Regression solution:

$$\beta = (X^T X + \lambda I)^{-1} X^T y$$

Step 1: Compute $X^T X$:

$$X^T X = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}^T \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix} = \begin{pmatrix} 5 & 5 \\ 5 & 10 \end{pmatrix}$$

Step 2: Add the regularization term λI :

$$X^T X + \lambda I = \begin{pmatrix} 5 & 5 \\ 5 & 10 \end{pmatrix} + 3 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 8 & 5 \\ 5 & 13 \end{pmatrix}$$

Step 3: Compute the inverse of $X^T X + \lambda I$:

$$(X^T X + \lambda I)^{-1} = \frac{1}{(8 \times 13) - (5 \times 5)} \begin{pmatrix} 13 & -5 \\ -5 & 8 \end{pmatrix} = \frac{1}{59} \begin{pmatrix} 13 & -5 \\ -5 & 8 \end{pmatrix}$$

Step 4: Compute $X^T y$:

$$X^T y = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}^T \begin{pmatrix} 36 \\ 62 \end{pmatrix} = \begin{pmatrix} 134 \\ 222 \end{pmatrix}$$

Step 5: Compute β :

$$\beta = \frac{1}{59} \begin{pmatrix} 13 & -5 \\ -5 & 8 \end{pmatrix} \begin{pmatrix} 134 \\ 222 \end{pmatrix} = \frac{1}{59} \begin{pmatrix} 632 \\ 1106 \end{pmatrix} = \begin{pmatrix} 10.71 \\ 18.75 \end{pmatrix}$$

Thus, the solution is:

$$\beta_1 \approx 10.71, \quad \beta_2 \approx 18.75$$